CNS Podcasts

OPEN

Maximizing Tumor Resection and Managing Cognitive Attentional Outcomes: Measures of Impact of Awake Surgery in Glioma Treatment

Luca Zigiotto, PsyD[‡§*], Riccardo Venturini, MSc[‡§*], Ludovico Coletta, PhD[||¶], Martina Venturini, MD[‡], Domenico Dal Monte, MSc[‡§], Laura Vavassori, MSc[‡¶], Francesco Corsini, MD[‡], Luciano Annicchiarico, MD[‡], Paolo Avesani, MSc[||¶], Costanza Papagno, MD, PhD[¶#], Silvio Sarubbo, MD, PhD[‡¶#]**

[‡]Department of Neurosurgery, "S. Chiara" University-Hospital, Azienda Provinciale per i Servizi Sanitari, Trento, Italy; [§]Department of Psychology, "S. Chiara" University-Hospital, Azienda Provinciale per i Servizi Sanitari, Trento, Italy; [||]Neuroinformatics Laboratory (NiLab), Fondazione Bruno Kessler, Trento, Italy; [¶]Center for Mind/Brain Sciences (CIMeC), University of Trento, Trento, Italy; [#]Centre for Medical Sciences (CISMED), University of Trento, Trento, Italy; **Department of Cellular, Computation and Integrative Biology (CIBIO), University of Trento, Trento, Italy

*Luca Zigiotto and Riccardo Venturini contributed equally to this work.

Correspondence: Luca Zigiotto, PsyD, Departments of Neurosurgery and Psychology, S. Chiara" University-Hospital, Azienda Provinciale per i Servizi Sanitari, Largo Medaglie D'Oro 9, Trento, TN 38100, Italy. Email: luca.zigiotto@apss.tn.it

Received, December 02, 2024; **Accepted,** April 08, 2025; **Published Online,** June 20, 2025.

Neurosurgery 98:365–375, 2026 https://doi.org/10.1227/neu.0000000000003591

Copyright © 2025 The Author(s). Published by Wolters Kluwer Health, Inc. on behalf of the Congress of Neurological Surgeons. This is an open access article distributed under the terms of the Creative Commons Attribution-Non Commercial-No Derivatives License 4.0 (CCBY-NC-ND), where it is permissible to download and share the work provided it is properly cited. The work cannot be changed in any way or used commercially without permission from the journal.

BACKGROUND AND OBJECTIVES: Patients with gliomas often experience neuropsychological deficits affecting their quality of life. Awake surgery (AwS) can reduce permanent cognitive deficits compared with asleep surgery (AsS), but it does not allow intraoperative mapping of all cognitive functions, including attention. Understanding how AwS and AsS affect attention is crucial, given its pivotal role in supporting various cognitive functions.

METHODS: We conducted a retrospective analysis on 64 glioma patients treated with AwS or AsS. Attention was assessed with visual search tasks and Trail Making Test Part A before and 1 week and 1 month after surgery. Volumetric T1-weighted and T2/Fluid Attenuated Inversion Recovery MRI sequences before and after surgery were used to delineate the lesion and the surgical cavity. The extent of resection was calculated to determine supramaximal resection for both contrast-enhanced and non–contrast-enhanced tumor regions.

RESULTS: There was a significant decrease in attentional scores 1 week after surgery, followed by a complete recovery. AwS was the only significant predictor of postoperative attentional deterioration. Univariate lesion analysis revealed negative association between lesions in the default mode network and postoperative attentional scores, whereas a multivariate network approach highlighted the involvement of several large-scale functional systems in sustaining attentional processes. AwS patients exhibited more extensive supramaximal resections of non–contrast-enhanced areas, which correlated with immediate postoperative attentional deterioration. The Kaplan-Meier analysis showed significantly longer overall survival for AwS patients with isocitrate dehydrogenase wild-type glioblastomas (mean days = 887.73) compared with AsS patients (mean days = 553.71; $P < .05$).

CONCLUSION: Although AwS enables a more extensive resection and thus an improved oncological outcome with longer overall survival rate, it also leads to higher transient postoperative decline in attentional performance. These results emphasize the need for careful patient selection, especially for lesions that involve anterior anatomical regions of the left default mode network. Future developments of standardized, reliable, and quantitative intraoperative monitoring of attention may further optimize surgical outcomes.

KEY WORDS: Asleep surgery, Attentional outcomes, Awake brain surgery, High-grade glioma, Low-grade glioma, Oncological outcomes, Overall survival

ABBREVIATIONS: ANOVA, Analysis of Variance; **AsS,** asleep surgery; **AwS,** awake surgery; **CE,** contrast-enhanced; **DMN,** default mode network; **EOR,** extent of resection; **FLAIR,** Fluid Attenuated Inversion Recovery; **FU,** follow-up; **HGGs,** high-grade gliomas; **IDH,** isocitrate dehydrogenase; **LGGs,** low-grade gliomas; **LNM,** lesion network mapping; **MGMT,** methylated-DNA-protein-cysteine methyltransferase; **OS,** overall survival; **RANO RG,** Response Assessment in Neuro-Oncology Resect Group; **RLSM,** region of interest-based lesion symptom mapping; **rs-fMRI,** functional resting-state MRI; **TMT-A,** Trail Making Test Part A.

Supplemental digital content is available for this article at neurosurgery-online.com.

CNS Journal Club Podcast and CME Exams available at cns.org/podcasts by scanning this QR code using your mobile device.

Prognosis of brain tumors varies widely depending on histology and biomolecular patterns[1,2] but generally remains poor for grade IV gliomas, with a median survival of 10 to 15 months.[1,3-5] Moreover, patients with gliomas frequently exhibit impaired neuropsychological profiles,[6-11] because of the tumor itself or treatment (ie, surgery and/or radiotherapy and chemotherapy).[12-14] Awake surgery (AwS) is therefore growingly adopted for both low-grade and high-grade gliomas (LGGs and HGGs, respectively), maximizing the extent of resection (EOR) and minimizing the risks of permanent deficits.[15-21] However, not all patients can undergo AwS and not all cognitive functions can be monitored during surgery,[22-25] such as attention.

Attention—defined as the "cognitive processes that enable the selection of, focus on, and sustained processing of information"[26]—is of particular interest because its interdependence with other cognitive domains and functions, especially the executive system,[27] may have significant and massive implications on patients' quality of life and on personal and professional activities.

Nevertheless, this crucial cognitive domain is not routinely monitored during AwS. In fact, to date, there is only one protocol for working memory,[28] which provides an incomplete picture, given the inherent and essential nature of the complex framework supporting attentional abilities: whole-brain, highly distributed, multi-interactive, and multimodal.[29-32]

Moreover, contradictory results on postoperative changes in attention are reported in the literature, with improvements[7,33,34] or impairments after surgery.[8,35-38] A more detailed understanding of mechanisms that influence the attentional domain is therefore crucial.

To this end, (1) we compared the attentional performance of patients with brain gliomas, who underwent either asleep surgery (AsS) or AwS; (2) we compared demographical and clinical data to explain and predict the outcome of attention abilities after surgery; (3) we analyzed oncological and survival outcomes of patients of AwS and AsS; and (4) we combined surgical resections with a functional resting-state MRI (rs-fMRI) atlas[39] and with normative rs-fMRI data,[40] providing new insights into the potential prediction of clinical outcomes.

METHODS

Population

A series of 64 consecutive patients (n = 64, 42 men, mean age 53.3 years, SD = ±13.5) were included in the study (62 right-handed, 1 left-handed, 1 ambidextrous, according to the Edinburgh Handedness Inventory Test[41]). Patients underwent awake (54.7%) or asleep (45.3%) surgery. Selection criteria are the following: no previous brain surgery, availability of presurgery and postsurgery MRI images, availability of neuropsychological assessment, and tumor gross-total resection. Regardless of tumor volume or location, AwS was proposed mainly based on patients' clinical, cognitive, and psychological status[42] (patients with good cooperative performance, no severe deficits in executive/language domains or anxiety disorders,[42-44] no anesthesiological contraindications[45]). Demographic and clinical characteristics are presented in Table 1 and reported in **Supplemental Digital Content 1** (http://links.lww.com/NEU/E857) with detailed inclusion criteria. This study was conducted according to the ethical standards of the Declaration of Helsinki and was approved by the Hospital Ethics Committee (Approval ID A734), for which patients gave their informed consent.

Neuropsychological Assessment

Each patient underwent a comprehensive neuropsychological assessment[20,34-36,46] of every cognitive domain 1 week before surgery ("*pre*"), 1 week after surgery ("*post*"), and 1 month after surgery ("*follow-up*"; FU).[20,35,36] Adjuvant treatments (ie, radio- and chemotherapy) for HGG were started within 6 weeks after surgery according to established neuro-oncological guidelines.[47] In this study that includes both LGG and HGG, we analyzed the 1-month FU, which was therefore collected before starting other therapies, to avoid potential negative effects on cognitive abilities. Furthermore, we only considered two attentional tasks as validated in the Italian population.[48,49] Description of both tests is reported in **Supplemental Digital Content 2** (http://links.lww.com/NEU/E858).

- *Attentional matrices* assess selective attention,[49] with higher scores indicating better performance
- *Trail Making Test Part A* (*TMT-A*)[48] measures visual scanning and attentional functions, with lower scores indicating better performance.

MRI Acquisition and Lesion Segmentation

The standard MRI evaluation protocol for glioma surgery includes the acquisition of structural, functional, and diffusion images. Details for magnetic resonance acquisition and lesion segmentation are reported in **Supplemental Digital Content 3** (http://links.lww.com/NEU/E859). Lesion segmentation was performed manually on the preoperative contrast-enhanced (CE) T1 and T2/Fluid Attenuated Inversion Recovery (FLAIR) images to define CE tumor volume and non-CE volume and on the early postoperative CE T1 and T2/FLAIR images acquired 24 hours after surgery to isolate the surgical cavity and non-CE residual volume. Figure 1 shows the distribution of surgical cavities. The lesion segmentation and the calculation of EOR were performed by an expert neurosurgeon blind to the patients' surgical group (AwS or AsS) and to the purpose of this study.

Region of Interest–Based Lesion Symptom Mapping Functional Lesion Analysis

RLSM[50] analysis was performed using NiiStat Toolbox for MATLAB (https://www.nitrc.org/projects/niistat/).[51,52] This analysis was performed using the Schaefer atlas, which takes into account 1000 parcels for 7 brain functional networks derived from rs-fMRI.[39] Specifications of region of interest–based lesion symptom mapping (RLSM) method are reported in **Supplemental Digital Content 4** (http://links.lww.com/NEU/E860).

Lesion Network Mapping Analysis

Recent evidence[53] suggests that standard univariate RLSM analysis may heavily underestimate the neural substrate associated with the emergence of specific behavioral deficits. To provide a more comprehensive description of the connectional landscape subtending attention, a lesion network mapping (LNM) analysis was performed,[40] with the same procedure outlined in Coletta et al[54]. Specifications of this analysis are reported in **Supplemental Digital Content 5** (http://links.lww.com/NEU/E861).

TABLE 1. Demographical Data and Distribution of Clinical Features for the Group of 64 Patients

Overall (N = 64)			
Mean age (SD)			53.3 y (±13.5)
Mean education (SD)			11.9 y (±3.74)
Mean surgical cavity volume (SD)			18.8 cc (±11.5)
Features	**Proportion (%)**	***P*-value**	**95% CI**
Tumor			
HGG	55 (85.9)	***P* < .001**	0.75-0.934
LGG	9 (14.1)		0.066-0.250
IDH mutation			
No	47 (73.4)	***P* < .001**	0.609-0.837
Yes	17 (26.6)		0.163-0.391
MGMT methylation			
No	44 (68.7)	***P* < .01**	0.559-0.798
Yes	20 (31.3)		0.202-0.441
Sex			
Female	22 (34.4)	***P* < .05**	0.229-0.473
Male	42 (65.6)		0.527-0.771
Side			
Right	29 (45.3)	*P* = .532	0.328-0.583
Left	35 (54.7)		0.417-0.672
Surgery			
Awake	35 (54.7)	*P* = .532	0.417-0.672
Asleep	29 (45.3)		0.328-0.583

HGG, high-grade glioma; IDH, Isocitrate dehydrogenase; LGG, low-grade glioma; MGMT, methylated-DNA-protein-cysteine methyltransferase.
The table includes results from proportion tests evaluating the distribution of clinical features in the entire cohort. 95% of CI is reported.

Surgical Procedure

Surgical resections were supported by a neuronavigation system that integrates structural images with tractography reconstructions of the critical white-matter pathways. During AwS, cortical and subcortical functional mapping was performed using a bipolar probe and direct electrical stimulation (60 Hz, 1 ms, 2-4 ranging mA) as previously described.[15-17,19-22] Specifications of awake monitoring are reported in **Supplemental Digital Content 6** (http://links.lww.com/NEU/E862). During AwS, it is possible to map or monitor specific functions,[20,22,28] improving the EOR.[20,55-57] However, some specific functions may only be clinically monitored: attentive functions were monitored indirectly during intraoperative neuropsychological testing through dual tasks (arm movement simultaneously performed with another cognitive task)[58,59] and assessment of executive functions by clinical observation and Stroop test.[60] Indeed, actually there are no established tools to monitor attentional functions. For AwS, tumor resection stopped when functional responses were elicited on different cognitive functions at cortical and subcortical stimulation, whereas for AsS, the resection stopped after the complete resection of the enhancing area and until reaching the reconstructions of the subcortical connections nearby the lesion.

Statistical Analysis

The analyses were conducted using SPSS v29.0, with significance level set at $P < .05$. Given the non-normal distribution (Shapiro-Wilk test <0.05), nonparametric analyses were performed. A binomial proportion test assessed demographical and clinical distributions. Friedman repeated-measures Analysis Of Variance (ANOVAs) were conducted for both tests, adjusted for age and education,[61] with preoperative, postoperative, and FU scores as factors. Post hoc analyses were performed using the paired Wilcoxon rank *t*-test. To quantify attentional decline after tumor resection, a delta (Δ) between the postoperative and preoperative scores (postoperative-preoperative) was calculated for both tests. Univariate linear regression examined the predictive effects of demographical and clinical characteristics on attentional decline. The EOR was calculated by comparing AsS and AwS using the nonparametric Mann-Whitney *U* test for independent samples. A two-tailed Spearman correlation assessed the relationship between EOR and Δ scores. RLSM was performed with a permutation threshold (10 000 permutations) to control for multiple comparisons,[62] using lesion volume as the regression coefficient.[62,63]

Factors influencing overall survival (OS) were assessed using a univariate Cox proportional hazard regression model. The Kaplan-Meier method with log-rank test assessed the difference in OS between AwS and AsS.[64] Considering the different glioma types in our cohort, OS analyses were performed only in patients with isocitrate dehydrogenase (IDH) wild-type glioblastoma (ie, grade 4 tumor without IDH mutation).[2] 95% CI, effect size (r) for significant results, and hazard ratio were reported for all analyses.

RESULTS

Demographical and Clinical Data

Our cohort was statistically unbalanced for HGG/LGG ratio ($P < .001$), IDH mutations ($P < .001$), methylated-DNA-protein-cysteine methyltransferase (MGMT) methylation ($P < .01$), and sex ($P < .05$). However, none of these characteristics influenced patients' attentional decline, as shown below. All results, including 95% CI, are reported in Table 1.

Attentional Scores Outcome

Friedman ANOVA showed a significant difference in attentional matrices ($P < .01$), with lower postoperative scores (mean accuracy = 68.5%, $P < .05$) compared with preoperative assessment (mean accuracy = 71.5%). FU scores showed an improvement (mean accuracy = 75%) compared with postoperative scores ($P < .01$). A similar result was observed for the TMT-A ($P < .01$), with worse postoperative scores (mean seconds = 48.2) compared with preoperative ones (mean seconds = 8.2, $P < .01$).

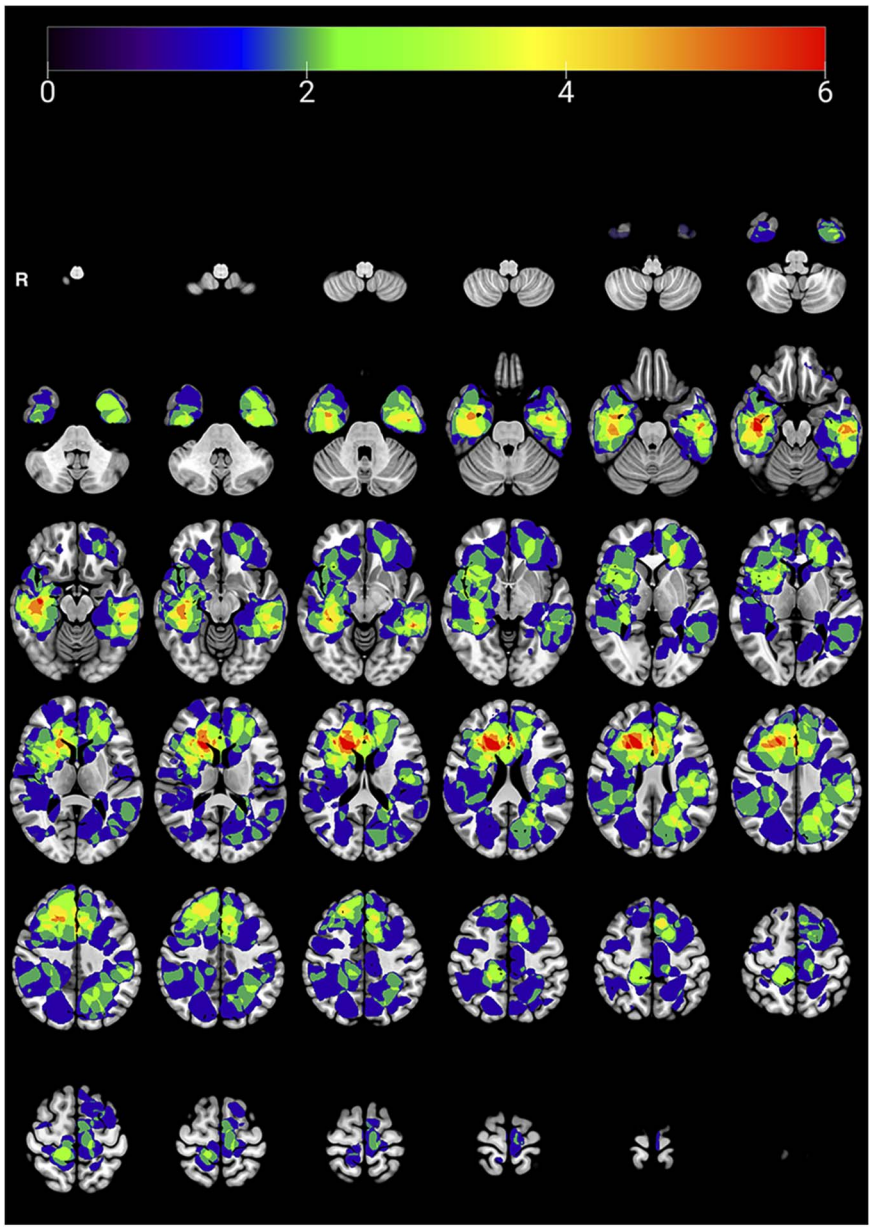

FIGURE 1. *Distribution and overlap of surgical cavities of the cohort of glioma patients. Images are in radiological convention (right hemisphere in the left side). R, right.*

Complete results are reported in Table 2 and Figure 2A, B. Paired Wilcoxon rank *T*-tests were performed separately for AwS and AsS; the results are reported in **Supplemental Digital Content 7** (http://links.lww.com/NEU/E863).

A Mann-Whitney *U* test was performed to compare neuropsychological scores of preoperative, postoperative, and FU separately by type of surgery: A significant difference emerged at the preoperative level, with higher baseline scores in the AwS group compared with the AsS group for both attentional matrices ($P < .05$) and TMT-A ($P < .05$). The results are reported in **Supplemental Digital Content 8** (http://links.lww.com/NEU/E864) and Figure 2C, D.

Factors Influencing Transitory Attentional Decline

We quantified the reduction in attentional scores after surgery using a Δ: patients had on average a −3.08% (SD = ±12.04%) reduction in stimuli detection at the attentional matrices test (Δ_{AM}) and an increase of +10.1 seconds

TABLE 2. Results of Friedman ANOVA, Comparing Trends in Attentional Performances in Attentional Matrices and TMT-A, and Results of Wilcoxon Test, Comparing Each Assessment Time Point

Test	Time	Mean score (SD), %	% of deficit	*P*-value Friedman (*P*-value Wilcoxon)
Attentional matrices	Pre	71.53 (±17.32)	14.06	χ^2 = 10.548; ***P* = .005**
	Post	68.46 (±18.72)	14.06	(pre vs post *Z* = −1.983; ***P* = .047**, *r* = −0.175)
	FU	74.97 (±14.22)	4.69	(post vs FU *Z* = −3.176; ***P* = .001**, *r* = −0.281) (pre vs FU *Z* = −1.627; *P* = .104)
TMT-A	Pre	38.17 (±24.04)	4.69	χ^2 = 9.325; ***P* = .009**
	Post	48.23 (±32.53)	12.5	(pre vs post *Z* = −2.849; ***P* = .004**, *r* = −0.252)
	FU	37.86 (±22.65)	1.6	(post vs FU *Z* = −2.991; ***P* = .003**, *r* = −0.264) (pre vs FU *Z* = −0.70; *P* = .944)

FU, follow-up; TMT-A, Trail Making Test part A.
For the Wilcoxon test, effect size (*r*) is reported alongside significance.

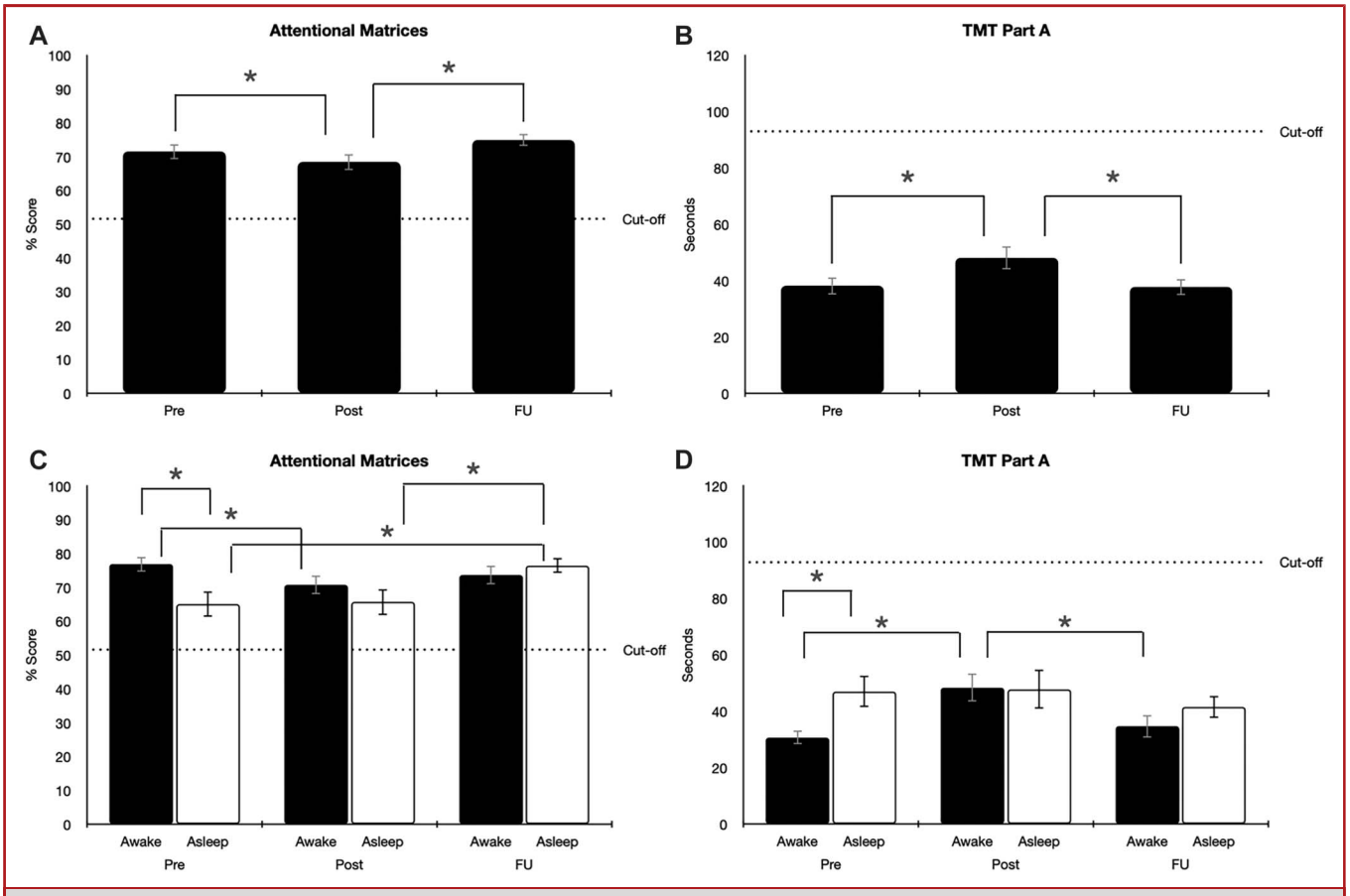

FIGURE 2. *Results of Friedman ANOVA on mean attentional scores (Attentional Matrices, TMT-A) at each time point (pre, post, FU), measured in percentage of correct answers for* **A**, *attentional matrices and* **B**, *in seconds taken to complete the task for TMT-A for the whole patient cohort, and* **C**, *attentional matrices and* **D**, *TMT-A for each type of surgical procedure. The cutoff was set at the threshold for the pathological score. Error bars represent standard errors of the mean. *P-value <.05. FU, follow-up; TMT-A, Trail Making Test Part A.*

TABLE 3. Results of the Univariate Linear Regression, Showing the Predictive Power of Demographical and Clinical Features in Explaining the Transitory Postoperative Attentional Worsening Compared With Preoperative Cognitive Assessment

	$\Delta_{(AM)}$				$\Delta_{(TMT)}$			
Factor	**B (95% CI)**	**SD**	**R^2**	***P*-value**	**B (95% CI)**	**SD**	**R^2**	***P*-value**
Sex	−2.12 (−5.91/+1.67)	1.89	0.02	*P* = .268	6 (−7.54/+19.54)	6.77	0.012	*P* = .379
Surgery	3.99 (+0.48/+7.51)	1.76	0.077	***P* = .026**	−17.01 (−29.28/−4.75)	6.14	0.11	***P* = .007**
Side	−1.27 (−4.91/+2.37)	1.82	0.008	*P* = .49	5.54 (−7.39/+18.46)	6.47	0.012	*P* = .395
Tumor	−2.28 (−7.48/+2.92)	2.6	0.012	*P* = .384	7.83 (−10.68/+26.34)	9.26	0.011	*P* = .401
IDH	2.61 (−1.45/+6.68)	2.03	0.026	*P* = .204	−10.89 (−25.28/+3.05)	7.2	0.036	*P* = .136
MGMT	−2.47 (−6.34/+1.41)	1.94	0.026	*P* = .207	3.36 (−10.58/+17.3)	6.97	0.004	*P* = .631
Location	−0.89 (−2.82/+1.05)	0.97	0.013	*P* = .364	0.3 (−6.63/+7.24)	3.47	0.000	*P* = .931
Age	−0.89 (−0.224/+0.04)	0.067	0.028	*P* = .189	0.31 (−0.17/+0.78)	0.24	0.026	*P* = .206

B, regression coefficient; IDH, isocitrate dehydrogenase; MGMT, methylated-DNA-protein-cysteine methyltransferase; TMT, Trail Making Test; R^2, coefficient of determination. Bold entries indicate statistically significant scores, with *P*-value <.05.

(SD = ±25.7) for completing the TMT-A ($\Delta_{TMT\text{-}A}$). There were no predictive effects of age, sex, tumor site or grade, IDH mutation, MGMT methylation, and tumor location on Δ_{AM} and $\Delta_{TMT\text{-}A}$ (all *P*s > 0.136). Interestingly, AsS or AwS was the only predictor for attentional decline after surgery (Δ_{AM} P < .05; $\Delta_{TMT\text{-}A}$ P < .01): Indeed, patients undergoing AwS showed a higher decrease of attentional scores for both Δ_{AM} (−6.09%) and $\Delta_{TMT\text{-}A}$ (+17.77 seconds) compared with AsS (+0.6% and +0.76 seconds, respectively). All results are reported in Table 3.

Non-CE EOR and Its Relationship With Postoperative Attentional Worsening

Although gross-total resection was achieved in all patients, Mann-Whitney *U* test determined that AwS allowed maximal supratotal resection of the non-CE area (mean 68.9%, SD = ±31.5%) compared with AsS (mean 42.7%, SD = ±33.5%; Z = −3.21, P < .01, r = −0.401). A two-tailed Spearman correlation confirmed a significant association between non-CE EOR and attention worsening, with a negative correlation for Δ_{AM} and a positive correlation with $\Delta_{TMT\text{-}A}$ (all *P*s < 0.05). As for control, we tested for similar results with preoperative CE and non-CE tumor volume, using postoperative surgical cavity and hyperintensity residual: importantly, no significant correlation emerged (all *P*s > 0.08). The results are reported in **Supplemental Digital Content 9** (http://links.lww.com/NEU/E865). Logistic regression showed that over 61% of non-CE EOR, there was a significantly attentional decline. The results are reported in **Supplemental Digital Content 10** (http://links.lww.com/NEU/E866) and **Supplemental Digital Content 11** (http://links.lww.com/NEU/E867).

RLSM Functional Lesion Analysis

The RLSM showed a significant negative association between lesions affecting the default mode network (DMN), particularly in the left prefrontal area, and postoperative attentional matrices scores (Z = −2.94, P < .05 after permutation correction) (Figure 3A). No significant negative Z-scores were found for the postoperative TMT-A scores, neither for Δ_{AM} nor for $\Delta_{TMT\text{-}A}$.

LNM Analysis

The LNM analysis revealed the involvement of a predominantly left-lateralized set of networks sustaining attentional processes. Of the 2918 voxels surviving the procedure and the selected threshold, 8.6% localized with somatomotor regions, 16.8% with the dorsal attention network, 56.9% with the ventral attention network, 15.6% with the frontoparietal network, and 1.9% with the DMN (Figure 3B).

Factors Influencing OS

From our original cohort, we separated 34 patients with IDH wild-type glioblastoma. Type of surgery (P < .05) and postoperative non-CE residual (P < .05) emerged as factors that could influence OS. Complete results are reported in **Supplemental Digital Content 12** (http://links.lww.com/NEU/E868). Non-CE residual was significantly lower in the AwS group (mean = 13.42 cc) compared with the AsS group (mean = 43.07 cc; Z = −2.32, P < .05). In addition, the Kaplan-Meier curve with log-rank test showed significantly longer OS rate for patients in the AwS group (mean days = 887.73; median days = 572; range 290-1893 days) compared with AsS (mean days = 553.71; median days = 440; range 175-1470 days P < .05). Full results are reported in Figure 4 and Table 4. Finally, Cox regression analysis showed that a non-CE residual volume <40 cc

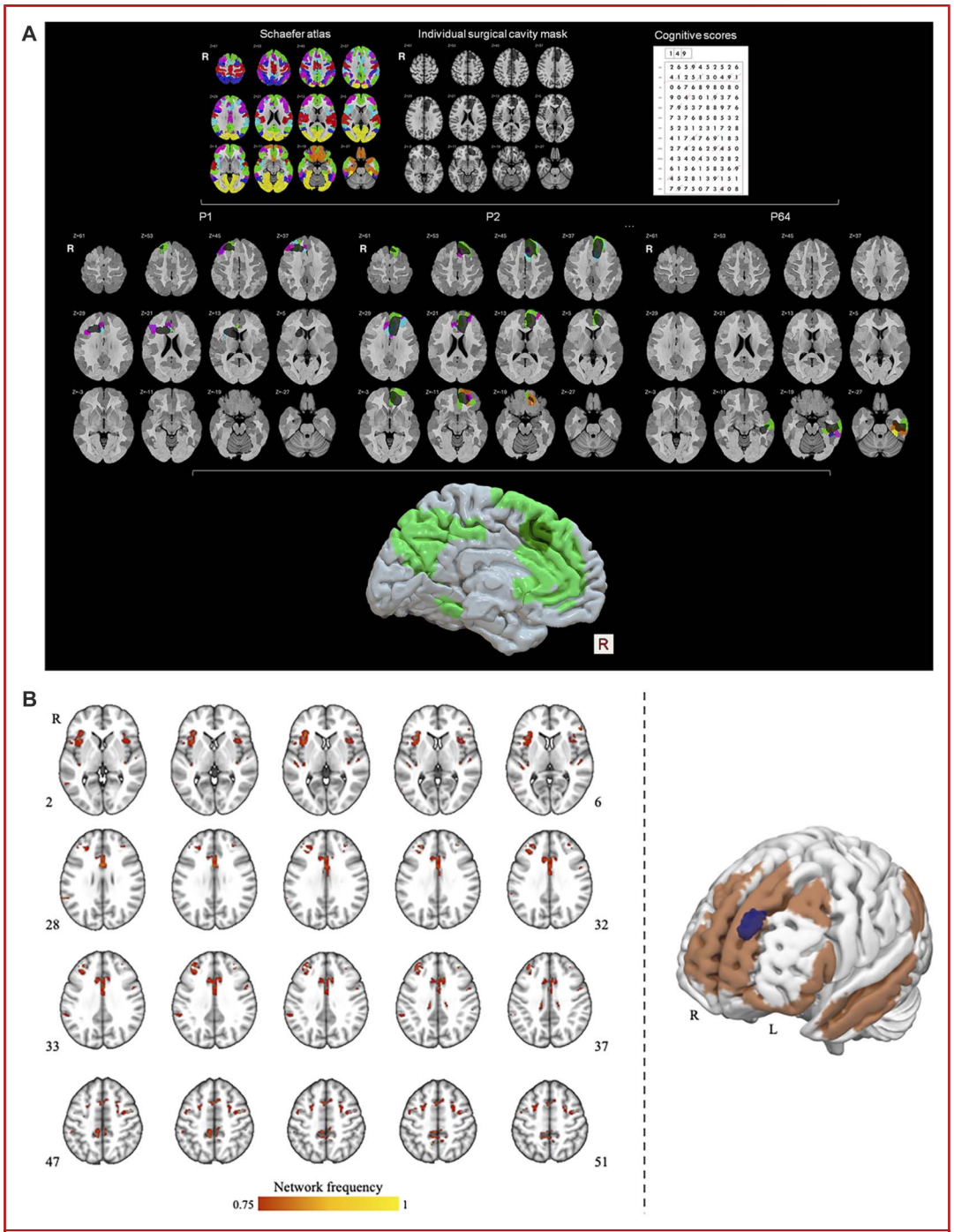

FIGURE 3. **A**, *Visual exemplary illustration of the data brought together by the RLSM functional lesion analysis. Top row: functional resting-state MRI atlas divided into 1000 parcels (Schaefer), example of surgical cavity (shaded) and example of attentional matrices. Middle row: intersection (colored) of example lesions (shaded) with networks parcellations from Schaefer atlas. Bottom row: the results of RLSM with negative statistically significant association between lesions in left prefrontal area (shaded) on the DMN (green) and attentional scores.* **B**, *The functional network obtained through lesion network mapping. The results of RLSM analysis with negative statistical significant association between lesions in left prefrontal area (blue) on the DMN (brown) and attentional scores. Images are in radiological convention (right hemisphere in the left side). DMN, default mode network; L, left; R, right; RLSM, region of interest-based lesion symptom mapping.*

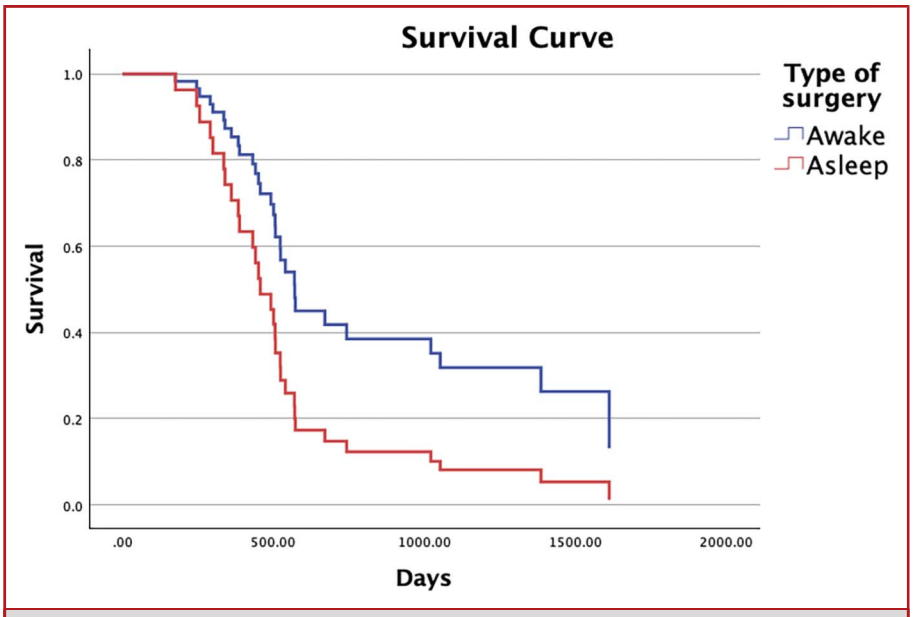

FIGURE 4. *Kaplan-Meier curves for patients belonging to the awake surgery group (in blue) and for patients belonging to the asleep surgery group (in red).*

predicted better survival rates ($P < .05$). The results are reported in **Supplemental Digital Content 13** (http://links.lww.com/NEU/E869) and **Supplemental Digital Content 14** (http://links.lww.com/NEU/E870).

DISCUSSION

To the best of our knowledge, this is the first retrospective analysis of the multiple variables that may influence the modulation of attentional performance after surgery, comparing 2 different surgical techniques (awake/asleep) and considering their different effects on the EOR and OS. Moreover, we compared the fluctuations in attentional performance considering the anatomofunctional substrate using consolidated rs-fMRI approaches.

Cognitive Outcome

In our cohort, a significant decline in attentional abilities was observed 1 week after surgery, which returned to preoperative levels after 1 month. Although previous studies documented cognitive fluctuations after brain surgery,[19,36,65,66] our findings are unique considering the comparison of AwS and AsS in a mixed series of brain gliomas, without confounding factors such as adjuvant treatments[47] that largely impair cognition.[67,68]

Considering all clinical variables (AsS or AwS, tumor grade, IDH mutation, MGMT status, laterality, and sex), AwS emerged as the only independent predictor of transient postoperative attentional impairment. A finer analysis showed that patients in the AwS group had a greater and significant decrease of attention immediately after surgery, with performance at 1 month comparable with preoperative performance. Conversely, the AsS group showed stable attention after surgery and an improvement 1 month after surgery for the attentional matrices. Interestingly, this result was observed only for patients of AwS, despite patients of AsS were submitted to longer general anesthesia which might itself cause fluctuations in the early postoperative outcome.[69] Moreover, despite similar attentional performance at both the 1-week and 1-month FU emerged, the preoperative assessments indicated that AwS patients had better attentional scores. This result is similar to data previously described[20] and reflects appropriate selection for AwS. Indeed, patients eligible for AwS must be cooperative, and severe deficits in attention and executive functions are exclusion criteria for this technique.[37,43] Therefore, as patients undergoing AwS show higher preoperative attentional performance, they are more likely to experience a transient decline postoperatively, with full recovery after 1 month. Despite this recovery (**Supplemental Digital Content 8**, http://links.lww.com/NEU/E864), attentional performance of AwS patients never improved with respect to the AsS patients. Therefore, patients' preoperative attentional abilities should also be considered when proposing AwS. However, the monitoring performed during AwS was enough to avoid permanent deficits. Attentional scores at 6-month and 12-month follow-up after surgery has been reported in **Supplemental Digital Content 15** (http://links.lww.com/NEU/E871).

Oncological Outcome

Increasing evidence reports that a low rate of residual FLAIR is critical for improving OS in HGGs. Recently, the Response Assessment in Neuro-Oncology (RANO) Resect Group (RANO RG) provided a new classification system for EOR in

TABLE 4. Data of Mean and Median Survival and Log-Rank Test

Patients	Group	N	Mean survival (d)	SE	Median survival (d)	SE	Log-rank test
Glioblastoma IDH-wildtype	Awake surgery	13	887.73	144.54	572	131.21	$\chi^2 = 4.12$ ***P* < .05**
	Asleep surgery	21	553.71	81.78	440	48.06	

IDH, isocitrate dehydrogenase. Bold entries indicate statistically significant scores, with *P*-value <.05.

glioblastomas based on the volume of non-CE residuals.[70] Nevertheless, EOR beyond the CE area in HGGs, which can be considered the interface between the core of the tumor and the healthy brain tissue,[71] may result in cognitive impairment more frequently than resection within the enhancing borders of the tumor. In our cohort, transient impairments in attentional abilities correlated exclusively with the supramaximal EOR achieved with AwS, resulting in higher rate of non-CE resection (68.9%) compared with AsS (42.7%). In addition, our cohort showed that 61% of non-CE EOR is the cutoff value predicting worsening of attentional capacity after surgery. To compare our results in this context, we performed survival analyses including only patients with IDH wild-type glioblastoma[1,2] (N = 34). When patients with the same tumor and biomolecular pattern (IDH-wildtype Glioblastoma) were considered, the most important factor for predicting OS was the volume of non-CE residuals, in line with the conclusions of RANO RG. Crucially, patients who underwent AwS had significantly lower residual volume (13.42 cc) and longer OS (887.73 days) than those who underwent AsS (43.07 cc; mean survival = 553.71 days). In-depth analyses showed that a non-CE residual volume smaller than 40 cc was the threshold that predicted higher OS. These results, even if limited by the sample size and the retrospective design, suggest that AwS can lead to less non-CE residual volume and therefore to higher survival rates. Finally, we computed the rate of patients' progression-free survival,[70,72] which was higher for AwS (mean = 537; median = 322; range 114-1893 days) compared with AsS (mean = 382; median = 296; range = 98-1381 days). Therefore, the selection of AwS technique should be tailored for each patient, considering the feasibility of an accurate mapping and monitoring with neuropsychological tools and the literature recommendations already reported, and also an expectable higher rate of transient deficits and the possible benefit regarding EOR and of oncological outcome.

Anatomofunctional Insights

Importantly, the results of LNM analysis corroborate the notion that attention encompasses several large-scale functional brain systems. We observed involvement of the left ventral attention network and of the dorsal anterior insula. Our results are in line with recent evidence,[73] suggesting that the anterior insula might act as gatekeeper of executive control—and thus also attention.

Our analysis showed a significant association between a worsening in selective attention immediately after surgery and lesions involving the DMN.[39] Although the DMN is primarily associated with internal thought processes,[74-76] its deactivation is essential for effective external goal-directed behavior.[77,78] Moreover, lesions to DMN may cause an ineffective suppression of this network which can impair selective attention, as its suppression is necessary for adaptive disengagement[73,79,80] and in attention-demanding tasks.[77-79] Indeed, DMN plays a crucial role in brain longitudinal functional reorganization after surgery.[81]

Limitations

The main limitations of this study are the small sample size, especially for survival analysis, and its retrospective design. In addition, patients' demographical and clinical characteristics were heterogeneous regarding sex, tumor grade, IDH mutation, and MGMT methylation.[1,2] However, none of these features proved to significantly affect postoperative attentional scores. Neuropsychological tests can be affected by cultural factors: Widely validated tests, such as the Trail Making Test, were used to minimize such biases. Finally, the restricted focus on attentional abilities gave us the possibility to reduce the number of statistical tests. Moreover, we implemented specific analyses to corroborate the robustness of the main results (**Supplemental Digital Content 7**, http://links.lww.com/NEU/E863). Notably, when the number of tests was increased, we carefully controlled for Type I errors with state-of-the-art approaches (10.000 permutations threshold for RLSM[62] and the threshold-free cluster enhancement approach for LNM[40,54]).

CONCLUSION

Among the clinical factors considered, the only one affecting the attentional outcome was AwS. This technique allowed a larger supramaximal EOR in HGGs, resulting in a temporary impairment at 1-week FU, while improving OS. One month after surgery, attentional scores recovered in both groups. Finally, we highlighted the role of a left prefrontal region in the DMN in attentional processes, providing new insights in the definition of safe functional margin of resection for attention.

Funding

This research was supported by the Grant No. 764/2021 of the Provincia Autonoma di Trento for the project NeuSurPlan.

Disclosures

The authors have no personal, financial, or institutional interest in any of the drugs, materials, or devices described in this article.

REFERENCES

1. Louis DN, Perry A, Reifenberger G, et al. The 2016 World Health Organization Classification of Tumors of the Central Nervous System: a summary. *Acta Neuropathol.* 2016;131(6):803-820.
2. Louis DN, Perry A, Wesseling P, et al. The 2021 WHO Classification of Tumors of the Central Nervous System: a summary. *Neuro Oncol.* 2021;23(8):1231-1251.
3. Luo C, Song K, Wu S, et al. The prognosis of glioblastoma: a large, multifactorial study. *Br J Neurosurg.* 2021;35(5):555-561.
4. Mohammed S, Dinesan M, Ajayakumar T. Survival and quality of life analysis in glioblastoma multiforme with adjuvant chemoradiotherapy: a retrospective study. *Rep Pract Oncol Radiother.* 2022;27(6):1026-1036.
5. Witthayanuwat S, Pesee M, Supaadirek C, Supakalin N, Thamronganantasakul K, Krusun S. Survival analysis of glioblastoma multiforme. *Asian Pac J Cancer Prev.* 2018;19(9):2613-2617.
6. Bergo E, Lombardi G, Guglieri I, Capovilla E, Pambuku A, Zagone V. Neurocognitive functions and health-related quality of life in glioblastoma patients: a concise review of the literature. *Eur J Cancer Care.* 2019;28(1):e12410.
7. Hoffermann M, Bruckmann L, Mahdy Ali K, Zaar K, Avian A, von Campe G. Pre- and postoperative neurocognitive deficits in brain tumor patients assessed by a computer based screening test. *J Clin Neurosci.* 2017;36:31-36.
8. Talacchi A, Santini B, Savazzi S, Gerosa M. Cognitive effects of tumour and surgical treatment in glioma patients. *J Neurooncol.* 2011;103(3):541-549.
9. Talacchi A, d'Avella D, Denaro L, et al. Cognitive outcome as part and parcel of clinical outcome in brain tumor surgery. *J Neurooncol.* 2012;108(2):327-332.
10. Van Kessel E, Baumfalk AE, Van Zandvoort MJE, Robe PA, Snijders TJ. Tumor-related neurocognitive dysfunction in patients with diffuse glioma: a systematic review of neurocognitive functioning prior to anti-tumor treatment. *J Neurooncol.* 2017;134(1):9-18.
11. Zucchella C, Bartolo M, Di Lorenzo C, Villani V, Pace A. Cognitive impairment in primary brain tumors outpatients: a prospective cross-sectional survey. *J Neurooncol.* 2013;112(3):455-460.
12. Aaronson NK, Taphoorn MJB, Heimans JJ, et al. Compromised health-related quality of life in patients with low-grade glioma. *J Clin Oncol.* 2011;29(33):4430-4435.
13. Hahn CA, Dunn RH, Logue PE, King JH, Edwards CL, Halperin EC. Prospective study of neuropsychologic testing and quality-of-life assessment of adults with primary malignant brain tumors. *Int J Radiat Oncol Biol Phys.* 2003;55(4):992-999.
14. Weitzner MA, Meyers CA, Byrne K. Psychosocial functioning and quality of life in patients with primary brain tumors. *J Neurosurg.* 1996;84(1):29-34.
15. Duffau H, Lopes M, Arthuis F, et al. Contribution of intraoperative electrical stimulations in surgery of low grade gliomas: a comparative study between two series without (1985-96) and with (1996-2003) functional mapping in the same institution. *J Neurol Neurosurg Psychiatry.* 2005;76(6):845-851.
16. Duffau H. Surgery of low-grade gliomas: towards a 'functional neurooncology'. *Curr Opin Oncol.* 2009;21(6):543-549.
17. Li YM, Suki D, Hess K, Sawaya R. The influence of maximum safe resection of glioblastoma on survival in 1229 patients: can we do better than gross-total resection? *J Neurosurg.* 2016;124(4):977-988.
18. Sarubbo S, Le Bars E, Moritz-Gasser S, Duffau H. Complete recovery after surgical resection of left Wernicke's area in awake patient: a brain stimulation and functional MRI study. *Neurosurg Rev.* 2012;35(2):287-292; discussion 292.
19. Van Kessel E, Snijders TJ, Baumfalk AE, et al. Neurocognitive changes after awake surgery in glioma patients: a retrospective cohort study. *J Neurooncol.* 2020;146(1):97-109.
20. Zigiotto L, Annicchiarico L, Corsini F, et al. Effects of supra-total resection in neurocognitive and oncological outcome of high-grade gliomas comparing asleep and awake surgery. *J Neurooncol.* 2020;148(1):97-108.
21. Lemaitre AL, Herbet G, Ng S, Moritz-Gasser S, Duffau H. Cognitive preservation following awake mapping-based neurosurgery for low-grade gliomas: a longitudinal, within-patient design study. *Neuro Oncol.* 2022;24(5):781-793.
22. Sarubbo S, Tate M, De Benedictis A, et al. Mapping critical cortical hubs and white matter pathways by direct electrical stimulation: an original functional atlas of the human brain. *Neuroimage.* 2020;205:116237.
23. Mandonnet E, Sarubbo S, Duffau H. Proposal of an optimized strategy for intraoperative testing of speech and language during awake mapping. *Neurosurg Rev.* 2017;40(1):29-35.
24. Papagno C. Studying cognitive functions by means of direct electrical stimulation: a review. *Neurol Sci.* 2017;38(12):2079-2087.
25. Sarubbo S, Annicchiarico L, Corsini F, et al. Planning brain tumor resection using a probabilistic atlas of cortical and subcortical structures critical for functional processing: a proof of concept. *Oper Neurosurg.* 2021;20(3):e175-e183.
26. Cohen RA. Attention. In: Kreutzer JS, DeLuca J, Caplan B, editors. *Encyclopedia of Clinical Neuropsychology*. Springer New York; 2011:282-292.
27. Ahissar M, Hochstein S. The spread of attention and learning in feature search: effects of target distribution and task difficulty. *Vis Res.* 2000;40(10-12):1349-1364.
28. Tomasino B, Guarracino I, Ius T, Skrap M. Continuous real-time neuropsychological testing during resection phase in left and right prefrontal brain tumors. *Curr Oncol.* 2023;30(2):2007-2020.
29. Corbetta M, Shulman GL. Control of goal-directed and stimulus-driven attention in the brain. *Nat Rev Neurosci.* 2002;3(3):201-215.
30. Gitelman DR, Nobre AC, Parrish TB, et al. A large-scale distributed network for covert spatial attention. *Brain.* 1999;122(6):1093-1106.
31. Petersen SE, Posner MI. The attention system of the human brain: 20 years after. *Annu Rev Neurosci.* 2012;35(1):73-89.
32. de Schotten MT, Dell'Acqua F, Forkel S, et al. A lateralized brain network for visuo-spatial attention. *Nat Neurosci.* 2011;14(10):1245-1246.
33. Ng JCH, See AAQ, Ang TY, Tan LYR, Ang BT, King NKK. Effects of surgery on neurocognitive function in patients with glioma: a meta-analysis of immediate post-operative and long-term follow-up neurocognitive outcomes. *J Neurooncol.* 2019;141(1):167-182.
34. Papagno C, Miracapillo C, Casarotti A, et al. What is the role of the uncinate fasciculus? Surgical removal and proper name retrieval. *Brain.* 2011;134(Pt 2):405-414.
35. Dallabona M, Sarubbo S, Merler S, et al. Impact of mass effect, tumor location, age, and surgery on the cognitive outcome of patients with high-grade gliomas: a longitudinal study. *Neurooncol Pract.* 2017;4(4):229-240.
36. Papagno C, Casarotti A, Comi A, Gallucci M, Riva M, Bello L. Measuring clinical outcomes in neuro-oncology. A battery to evaluate low-grade gliomas (LGG). *J Neurooncol.* 2012;108(2):269-275.
37. Santini B, Talacchi A, Squintani G, Casagrande F, Capasso R, Miceli G. Cognitive outcome after awake surgery for tumors in language areas. *J Neurooncol.* 2012;108(2):319-326.
38. Tabor JK, Bonda D, LeMonda BC, D'Amico RS. Neuropsychological outcomes following supratotal resection for high-grade glioma: a review. *J Neurooncol.* 2021;152(3):429-437.
39. Schaefer A, Kong R, Gordon EM, et al. Local-global parcellation of the human cerebral cortex from intrinsic functional connectivity MRI. *Cereb Cortex.* 2018;28(9):3095-3114.
40. Fox MD. Mapping symptoms to brain networks with the human connectome. *N Engl J Med.* 2018;379(23):2237-2245.
41. Oldfield RC. The assessment and analysis of handedness: the Edinburgh inventory. *Neuropsychologia.* 1971;9(1):97-113.
42. Seidel K, Szelényi A, Bello L. Intraoperative mapping and monitoring during brain tumor surgeries. *Handb Clin Neurol.* 2022;186:133-149.
43. Nossek E, Matot I, Shahar T, et al. Failed awake craniotomy: a retrospective analysis in 424 patients undergoing craniotomy for brain tumor. *J Neurosurg.* 2013;118(2):243-249.
44. Santini B, Talacchi A, Casagrande F, et al. Eligibility criteria and psychological profiles in patient candidates for awake craniotomy: a pilot study. *J Neurosurg Anesthesiol.* 2012;24(3):209-216.
45. Cormack JR, Costello TG. Awake craniotomy: anaesthetic guidelines and recent advances. *J Clin Neurosci.* 2004;11(1):16–19.
46. Zigiotto L, Vavassori L, Annicchiarico L, et al. Segregated circuits for phonemic and semantic fluency: a novel patient-tailored disconnection study. *Neuroimage Clin.* 2022;36:103149.
47. Stupp R, Mason WP, van den Bent MJ, et al. Radiotherapy plus concomitant and adjuvant temozolomide for glioblastoma. *N Engl J Med.* 2005;352(10):987-996.
48. Giovagnoli AR, Del Pesce M, Mascheroni S, Simoncelli M, Laiacona M, Capitani E. Trail making test: normative values from 287 normal adult controls. *Ital J Neurol Sci.* 1996;17(4):305-309.
49. Spinnler H, Tognoni G. Taratura e standardizzazione italiana di test neuropsicologici. *Ital J Neurol Sci.* 1987;8(Suppl 8):8-120.
50. Fridriksson J, Den Ouden DB, Hillis AE, et al. Anatomy of aphasia revisited. *Brain.* 2018;141(3):848-862.

51. Hoeren M, Kümmerer D, Bormann T, et al. Neural bases of imitation and pantomime in acute stroke patients: distinct streams for praxis. *Brain.* 2014;137(Pt 10):2796-2810.
52. Smith DV, Clithero JA, Rorden C, Karnath HO. Decoding the anatomical network of spatial attention. *Proc Natl Acad Sci USA.* 2013;110(4):1518-1523.
53. Joutsa J, Moussawi K, Siddiqi SH, et al. Brain lesions disrupting addiction map to a common human brain circuit. *Nat Med.* 2022;28(6):1249-1255.
54. Coletta L, Avesani P, Zigiotto L, et al. Integrating direct electrical brain stimulation with the human connectome. *Brain.* 2024;147(3):1100-1111.
55. Yordanova YN, Moritz-Gasser S, Duffau H. Awake surgery for WHO Grade II gliomas within "noneloquent" areas in the left dominant hemisphere: toward a "supratotal" resection. Clinical article. *J Neurosurg.* 2011;115(2):232-239.
56. Motomura K, Ohka F, Aoki K, Saito R. Supratotal resection of gliomas with awake brain mapping: maximal tumor resection preserving motor, language, and neurocognitive functions. *Front Neurol.* 2022;13:874826.
57. Duffau H. New philosophy, clinical pearls, and methods for intraoperative cognition mapping and monitoring "à la carte" in brain tumor patients. *Neurosurgery.* 2021;88(5):919-930.
58. Duffau H, Ng S, Lemaitre AL, Moritz-Gasser S, Herbet G. Constant multi-tasking with time constraint to preserve across-network dynamics throughout awake surgery for low-grade glioma: a necessary step to enable patients resuming an active life. *Front Oncol.* 2022;12:924762.
59. Duffau H. Indications of awake mapping and selection of intraoperative tasks. In: *Brain Mapping: From Neural Basis of Cognition to Surgical Applications.* Springer; 2011:321-334.
60. Puglisi G, Howells H, Sciortino T, et al. Frontal pathways in cognitive control: direct evidence from intraoperative stimulation and diffusion tractography. *Brain.* 2019;142(8):2451-2465.
61. Capitani E, Laiacona M. Aging and psychometric diagnosis of intellectual impairment: some considerations on test scores and their use. *Dev Neuropsychol.* 1988; 4(4):325-330.
62. Karnath HO, Sperber C, Rorden C. Mapping human brain lesions and their functional consequences. *NeuroImage.* 2018;165:180-189.
63. Sperber C, Karnath HO. Impact of correction factors in human brain lesion-behavior inference. *Hum Brain Mapp.* 2017;38(3):1692-1701.
64. Kaplan EL, Meier P. Nonparametric estimation from incomplete observations. *J Am Stat Assoc.* 1958;53(282):457-481.
65. Duffau H, Capelle L, Denvil D, et al. Functional recovery after surgical resection of low grade gliomas in eloquent brain: hypothesis of brain compensation. *J Neurol Neurosurg Psychiatry.* 2003;74(7):901-907.
66. Ng S, Herbet G, Lemaitre AL, Cochereau J, Moritz-Gasser S, Duffau H. Neuropsychological assessments before and after awake surgery for incidental low-grade gliomas. *J Neurosurg.* 2021;135(3):871-880.
67. Kirkman MA, Hunn BHM, Thomas MSC, Tolmie AK. Influences on cognitive outcomes in adult patients with gliomas: a systematic review. *Front Oncol.* 2022;12:943600.
68. Wefel JS, Kayl AE, Meyers CA. Neuropsychological dysfunction associated with cancer and cancer therapies: a conceptual review of an emerging target. *Br J Cancer.* 2004;90(9):1691-1696.
69. Hem S, Albite R, Loresi M, et al. Pathological changes of the hippocampus and cognitive dysfunction following frontal lobe surgery in a rat model. *Acta Neurochir (Wien).* 2016;158(11):2163-2171.
70. Karschnia P, Young JS, Dono A, et al. Prognostic validation of a new classification system for extent of resection in glioblastoma: a report of the RANO resect group. *Neuro Oncol.* 2023;25(5):940-954.
71. Silva M, Vivancos C, Duffau H. The concept of «Peritumoral Zone» in diffuse low-grade gliomas: oncological and functional implications for a connectome-guided therapeutic attitude. *Brain Sci.* 2022;12(4):504.
72. Han K, Ren M, Wick W, et al. Progression-free survival as a surrogate endpoint for overall survival in glioblastoma: a literature-based meta-analysis from 91 trials. *Neuro Oncol.* 2014;16(5):696-706.
73. Molnar-Szakacs I, Uddin LQ. Anterior insula as a gatekeeper of executive control. *Neurosci Biobehav Rev.* 2022;139:104736.
74. Fox MD, Snyder AZ, Vincent JL, Corbetta M, Van Essen DC, Raichle ME. The human brain is intrinsically organized into dynamic, anticorrelated functional networks. *Proc Natl Acad Sci USA.* 2005;102(27):9673-9678.
75. Greicius MD, Krasnow B, Reiss AL, Menon V. Functional connectivity in the resting brain: a network analysis of the default mode hypothesis. *Proc Natl Acad Sci USA.* 2003;100(1):253-258.
76. Raichle ME, MacLeod AM, Snyder AZ, Powers WJ, Gusnard DA, Shulman GL. A default mode of brain function. *Proc Natl Acad Sci USA.* 2001;98(2):676-682.
77. Menon V. 20 years of the default mode network: a review and synthesis. *Neuron.* 2023;111(16):2469-2487.
78. Weissman DH, Roberts KC, Visscher KM, Woldorff MG. The neural bases of momentary lapses in attention. *Nat Neurosci.* 2006;9(7):971-978.
79. Menon V, Uddin LQ. Saliency, switching, attention and control: a network model of insula function. *Brain Struct Funct.* 2010;214(5-6):655-667.
80. Sridharan D, Levitin DJ, Menon V. A critical role for the right fronto-insular cortex in switching between central-executive and default-mode networks. *Proc Natl Acad Sci USA.* 2008;105(34):12569-12574.
81. Saviola F, Zigiotto L, Novello L, et al. The role of the default mode network in longitudinal functional brain reorganization of brain gliomas. *Brain Struct Funct.* 2022;227(9):2923-2937.

Acknowledgments

Authors thank the Health Department of Provincia Autonoma di Trento and the Direction Team of the Azienda Provinciale per i Servizi Sanitari, for their support to clinical and research activities of the Neurosurgical Department. Thanks also to DZ for his support. Author Contribution: Conceptualization: L.Z. and S.S.; Data collection: L.Z., L.A., F.C., R.V., M.V., D.D.M., S.S.; Methodology: L.Z., R.V., M.V., D.D.M., L.V., L.C., P.A., C.P., S.S.; Data analysis: L.Z., R.V., L.C.; Visualization: L.Z., R.V., D.D.M., L.V., L.C., P.A., C.P., S.S.; Writing-original draft: L.Z., R.V., and S.S.; All authors reviewed the manuscript.

Supplemental digital content *is available for this article at neurosurgery-online.com.*

Supplemental Digital Content 1. Patients' demographical and clinical characteristics and inclusion criteria.
Supplemental Digital Content 2. Description of attentional matrices test and Trail Making Test.
Supplemental Digital Content 3. Details of MR acquisition and lesion segmentation.
Supplemental Digital Content 4. Specifications of ROI-based lesion symptom mapping (RLSM) method.
Supplemental Digital Content 5. Specifications of lesion network mapping (LNM) method.
Supplemental Digital Content 6. Specifications of awake monitoring.
Supplemental Digital Content 7. Results of paired Wilcoxon rank T-tests performed separately for AwS and AsS.
Supplemental Digital Content 8. Table 1. Reporting results of Mann-Whitney *U* test performed to compare neuropsychological scores of preoperative, postoperative, and FU separately by type of surgery.
Supplemental Digital Content 9. Table 2. Reporting results of two-tailed Spearman correlation between attention worsening after surgery and postoperative non-CE EOR, preoperative CE, and non-CE tumor volume.
Supplemental Digital Content 10. Figure 1. Reporting logistic regression. This analysis showed that over 61% of non-CE EOR, there was a significantly attentional decline.
Supplemental Digital Content 11. Text describing results of logistic regression. This analysis showed that over 61% of non-CE EOR, there was a significantly attentional decline.
Supplemental Digital Content 12. Table 3. Reporting results of univariate Cox proportional hazard regression model. 95% CI = 95% of confidence interval on hazard ratio.
Supplemental Digital Content 13. Figure 2. Reporting Cox regression. This analysis showed that a non-CE residual volume <40 cc predicted better survival rates.
Supplemental Digital Content 14. Text reporting Cox regression. This analysis showed that a non-CE residual volume <40 cc predicted better survival rates.
Supplemental Digital Content 15. Table 4. Reporting neuropsychological scores of preoperative, postoperative, and 1-month, 6-month, and 12-month FU separately by type of surgery.

CNS Journal Club Podcast *and CME Exams available at cns.org/podcasts.*